

DA 3014 – Statistics Laboratory III

Assignment 3

Prediction of 30-Day Hospital Readmission

Among Patients with Diabetes

Dataset:

Diabetes 130-US Hospitals for Years 1999–2008

Index Number: 24ada006

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Table of Contents

1. **Introduction and Objective**
 - 1.1 Analytical Objectives
2. **Dataset Selection and Description**
 - 2.1 Dataset Selection
 - 2.2 Response Variable
 - 2.3 Predictor Variables
 - 2.4 Suitability for the Assignment
3. **Data Cleaning and Preprocessing**
 - 3.1 Initial Data Quality Assessment
 - 3.2 Removal of Identifier and Leakage Variables
 - 3.3 Grouping of High-Cardinality Categorical Variables
 - 3.4 Grouping of Medical Specialty
 - 3.5 Grouping of Admission, Discharge, Source Validity
 - 3.6 Treatment of Rare Categories
 - 3.7 Medication Variables
 - 3.8 Final Modelling Dataset
 - 3.9 Training and Test Split
4. **Exploratory Data Analysis**
 - 4.1 Numerical Variable Distributions
 - 4.2 Categorical Variable Distributions
 - 4.3 Age and 30-Day Readmission
 - 4.4 Previous Inpatient Visits and Readmission
 - 4.5 Previous Emergency Visits and Readmission
 - 4.6 Length of Stay and Number of Medications
 - 4.7 Diagnosis Group and Insulin Status
 - 4.8 Outliers and Unusual Patterns
 - 4.9 Expected Important Predictors
5. **Logistic Regression Analysis**
 - 5.1 Full Binary Logistic Regression Model
 - 5.2 Most Statistically Significant Predictors
 - 5.3 Odds Ratio Interpretation
 - 5.4 Reduced Logistic Regression Model
 - 5.5 Full and Reduced Model Performance
 - 5.6 Effect of the Classification Threshold
 - 5.7 Confusion Matrix Interpretation
 - 5.8 Logistic Regression Assessment
6. **Machine-Learning Models**
 - 6.1 Random Forest Classifier
 - 6.2 Random Forest Hyperparameter Tuning
 - 6.3 Tuned Random Forest Performance
 - 6.4 Random Forest Overfitting Assessment

- 6.5 Random Forest Variable Importance
- 6.6 Gradient Boosting Classifier
- 6.7 Initial Gradient Boosting Overfitting Assessment
- 6.8 Gradient Boosting Hyperparameter Tuning
- 6.9 Tuned Gradient Boosting Performance
- 6.10 Tuned Gradient Boosting Overfitting Assessment
- 6.11 Machine-Learning Model Summary
- 7. **Model Comparison and Discussion**
 - 7.1 Comparison of Final Models
 - 7.2 Recommended Model
 - 7.3 Comparison with Exploratory Data Analysis
 - 7.4 Comparison of Important Predictors Across Models
 - 7.5 Effect of Classification Threshold
 - 7.6 Overall Discussion
- 8. **Limitations**
- 9. **Conclusion**
- 10. **References and Data Source**
- 11. **Code Availability**

1. Introduction and Objective

Diabetes is a chronic condition that frequently requires hospital care, and repeated hospital admissions can place a considerable burden on both patients and healthcare systems. Identifying patients who are at higher risk of early readmission may therefore support better discharge planning, follow-up care, and allocation of healthcare resources.

This study uses the **Diabetes 130-US Hospitals for Years 1999–2008** dataset obtained from the UCI Machine Learning Repository. The dataset contains hospital encounter information for patients with diabetes across multiple hospitals in the United States.

The primary objective of this analysis is to develop and compare statistical and machine-learning models for predicting whether a patient will be **readmitted to hospital within 30 days** following discharge.

The original readmission variable contains three categories:

- <30 – readmitted within 30 days
- >30 – readmitted after more than 30 days
- NO – no recorded readmission

For this analysis, a binary response variable named `readmit_30` was created:

1 = Readmitted within 30 days

0 = No readmission within 30 days

or readmission after more than 30 days

The analysis includes exploratory data analysis, logistic regression, Random Forest, and Gradient Boosting models. Their predictive performance is compared using measures appropriate for an imbalanced binary classification problem, including precision, recall, F1-score, and ROC-AUC.

1.1 Analytical Objectives

The specific objectives are to:

1. clean and preprocess the hospital encounter data;
2. investigate the distributions and relationships of potential predictors with 30-day readmission;
3. identify variables that may be associated with early readmission;
4. develop a full and reduced binary logistic regression model;

5. develop and tune Random Forest and Gradient Boosting classifiers;
6. compare the predictive performance of the statistical and machine-learning approaches; and
7. examine how changing the classification threshold affects the identification of patients at risk of readmission.

2. Dataset Selection and Description

2.1 Dataset Selection

The dataset selected for this study is **Diabetes 130-US Hospitals for Years 1999–2008**, obtained from the **UCI Machine Learning Repository**.

The dataset contains hospital encounter records for patients with diabetes from 130 hospitals and integrated delivery networks in the United States over the period 1999 to 2008. Each row represents a hospital encounter rather than a unique patient, meaning that the same patient may appear more than once if they had multiple admissions.

The dataset was selected because it satisfies the main requirements of the assignment. It is a real-world healthcare dataset, contains more than 1,000 observations, includes both numerical and categorical predictors, and supports a binary classification problem. The dataset contains **101,766 observations** and **50 columns** in the downloaded CSV file.

2.2 Response Variable

The original response variable, readmitted, contains three categories:

NO = No recorded readmission

>30 = Readmitted after more than 30 days

<30 = Readmitted within 30 days

For this study, the response was converted into a binary variable named readmit_30:

1 = Readmitted within 30 days

0 = No readmission or readmission after more than 30 days

Readmission category	Number of encounters	Percentage
----------------------	----------------------	------------

NO	54,864	53.91%
>30	35,545	34.93%
<30	11,357	11.16%

After conversion to the binary response variable:

readmit_30	Number of encounters	Percentage
------------	----------------------	------------

0	90,409	88.84%
1	11,357	11.16%

This shows that the response variable is **imbalanced**, with substantially fewer patients readmitted within 30 days. Therefore, later model evaluation considers measures such as recall, F1-score, and ROC-AUC in addition to overall accuracy

2.3 Predictor Variables

The dataset contains a mixture of numerical and categorical variables relating to patient characteristics, hospital utilisation, diagnoses, treatments, medications, and admission or discharge information.

The main numerical variables considered include:

- time_in_hospital
- num_lab_procedures
- num_procedures
- num_medications
- number_outpatient
- number_emergency
- number_inpatient
- number_diagnoses

Categorical variables include patient demographics, medication status, diagnosis groups, medical specialty, admission type, discharge disposition, and admission source.

Several variables required preprocessing before modelling because of missing values, high-cardinality categories, coded identifiers, or extremely rare levels. These steps are described in the next section.

2.4 Suitability for the Assignment

The dataset was considered suitable because it:

- represents a real-world healthcare classification problem;
- contains a binary outcome suitable for logistic regression and machine-learning classification;
- includes both numerical and categorical predictors;
- contains substantially more than 1,000 observations;
- allows investigation of class imbalance, feature importance, and predictive performance;
- supports comparison between statistical and machine-learning approaches.

3. Data Cleaning and Preprocessing

3.1 Initial Data Quality Assessment

The original dataset contained **101,766 hospital encounters and 50 columns**. An initial assessment was conducted to identify missing values, identifier variables, high-cardinality categorical variables, and variables with extremely sparse information.

Several variables contained very high proportions of missing values. In particular:

Variable	Missing values	Missing percentage
weight	98,569	96.86%
max_glu_serum	96,420	94.75%
A1Cresult	84,748	83.28%
medical_specialty	49,949	49.08%
payer_code	40,256	39.56%

Because weight, max_glu_serum, and A1Cresult contained more than 80% missing values, they were removed from the modelling dataset. payer_code was also removed because of its high level of missingness and its administrative nature.

The missing values in retained categorical variables such as race, medical_specialty, and the diagnosis fields were represented using an Unknown category rather than deleting the corresponding observations.

3.2 Removal of Identifier and Leakage Variables

The variables `encounter_id` and `patient_nbr` were removed because they are identifiers rather than meaningful predictive features.

The original `readmitted` variable was also removed after creating the binary response `readmit_30`. Retaining the original variable would have caused target leakage because it directly contains the information used to construct the response.

3.3 Grouping of High-Cardinality Categorical Variables

Several categorical variables contained a large number of distinct categories, which would have produced a very large number of dummy variables and reduced model interpretability. These variables were therefore grouped into broader categories before modelling.

The diagnosis variables `diag_1`, `diag_2`, and `diag_3` contained many individual diagnosis codes. These codes were grouped into the following broader categories:

- Circulatory
- Respiratory
- Digestive
- Diabetes
- Injury
- Musculoskeletal
- Genitourinary
- Neoplasms
- Other
- Unknown

After creating `diag_1_group`, `diag_2_group`, and `diag_3_group`, the original diagnosis-code variables were removed.

The `medical_specialty` variable also contained a large number of categories, with 73 distinct specialties. Specialties with at least 1,000 observations were retained as separate categories, while less frequent specialties were combined into an `Other` category. Missing medical-specialty values were retained as `Unknown`.

These grouping steps reduced dimensionality while preserving clinically and operationally meaningful information for the subsequent statistical and machine-learning models.

3.4 Grouping of Medical Specialty

medical_specialty contained 73 categories, including many with very few observations. Categories with at least 1,000 observations were retained individually, while less frequent categories were combined into an Other group.

Missing medical-specialty values were retained as a separate Unknown category

3.5 Grouping of Admission, Discharge, and Source Variables

The coded variables admission_type_id, discharge_disposition_id, and admission_source_id were converted into broader interpretable categories using the accompanying mapping file.

The resulting grouped variables were:

- admission_type_group
- discharge_group
- admission_source_group

The original coded identifier variables were then removed.

Patients classified as Expired in the discharge group were excluded from the modelling population because they were not at risk of subsequent hospital readmission.

3.6 Treatment of Rare Categories

Extremely rare categorical levels created instability during logistic-regression estimation. Therefore, selected rare categories were combined into broader groups.

For example:

- rare admission-source categories were combined into Unknown;
- diag_1_group = Unknown was combined into Other;
- admission_type_group = Other was combined into Unknown;
- three observations with gender = Unknown/Invalid were removed.

These adjustments allowed the final logistic regression model to converge successfully.

3.7 Medication Variables

Medication variables with extremely low frequencies were removed because they contributed very little information and could create unstable estimates.

The retained medication variables included commonly used treatments such as metformin, glimepiride, glipizide, glyburide, pioglitazone, rosiglitazone, repaglinide, and insulin.

3.8 Final Modelling Dataset

After cleaning and preprocessing, the final modelling dataset contained **28 predictors and one binary response variable**.

Categorical predictors were converted to numerical dummy variables using one-hot encoding. One category from each categorical predictor was omitted as the reference category to avoid perfect multicollinearity.

After the final rare-category adjustments, the encoded modelling dataset contained **96 predictor terms**.

3.9 Training and Test Split

The dataset was divided into training and test sets using an **80:20 stratified split** with `random_state = 42`.

After excluding expired encounters and completing the final preprocessing steps, the modelling samples contained approximately:

Training set: 80,074 observations

Test set: 20,037 observations

Stratification was used so that the proportion of 30-day readmissions remained similar in both the training and test datasets.

4. Exploratory Data Analysis

4.1 Numerical Variable Distributions

Exploratory analysis was first carried out on the main numerical predictors:

- time_in_hospital
- num_lab_procedures
- num_procedures
- num_medications
- number_outpatient
- number_emergency
- number_inpatient
- number_diagnoses

Summary statistics and histograms were used to examine their distributions.

The average hospital stay was approximately **4.40 days**, with a median of 4 days and a maximum of 14 days. The number of laboratory procedures averaged approximately **43 per encounter**, while the average number of medications was approximately **16**.

Several healthcare-utilisation variables, particularly number_outpatient, number_emergency, and number_inpatient, were strongly right-skewed. Most encounters involved patients with few or no previous visits, while a relatively small number had much higher utilisation.

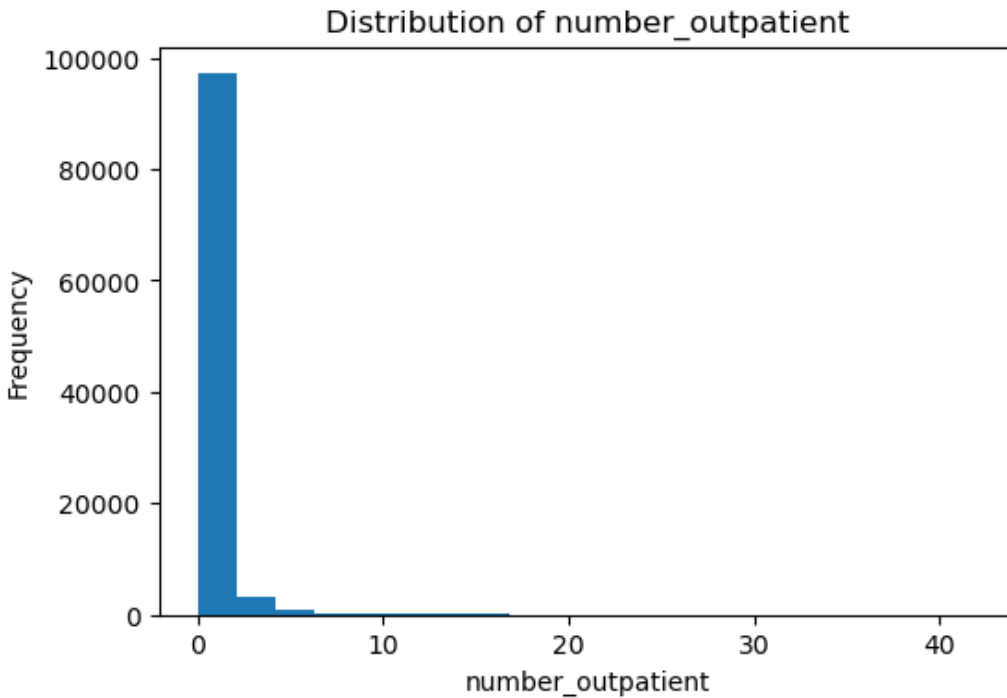


Figure 1

The distribution of `number_outpatient` is strongly right-skewed. Most hospital encounters involved patients with no or very few previous outpatient visits, while a small number of patients had substantially higher outpatient utilisation. The extreme values were retained because they may represent genuine high-utilisation patients and could contain useful predictive information.

4.2 Categorical Variable Distributions

The distributions of key categorical predictors were examined using frequency tables and bar charts.

The dataset contained slightly more female than male encounters. The largest age groups were patients aged **60–80 years**, particularly the [70-80) category.

Other categorical variables examined included race, admission type, discharge disposition, admission source, primary diagnosis group, and medical specialty group.

Because the report is intended to summarise the main findings, only selected categorical plots are included, while the complete set remains available in the Python notebook

4.3 Age and 30-Day Readmission

Readmission rates were examined across age groups.

The observed 30-day readmission rate generally varied between approximately **10% and 12%** for many adult age groups, although the pattern was not strictly increasing with age.

For example:

- [50-60) — approximately 9.7%
- [60-70) — approximately 11.1%
- [70-80) — approximately 11.8%
- [80-90) — approximately 12.1%

The youngest age groups contained relatively few observations, so their readmission rates were interpreted cautiously.

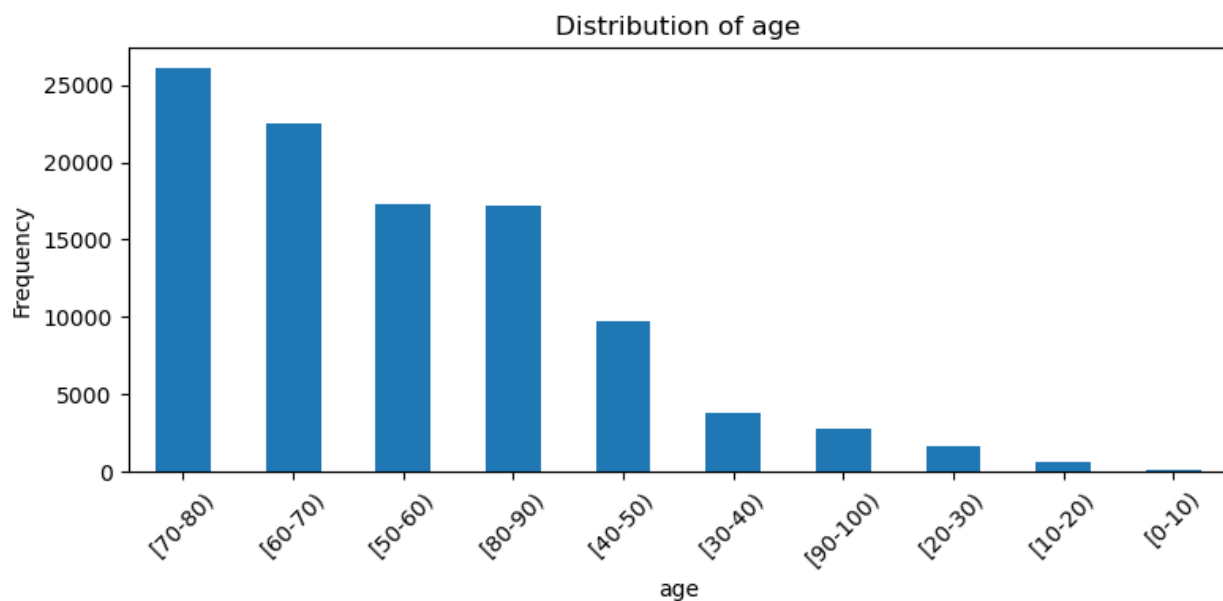


Figure 2

The age distribution shows that most hospital encounters involved older patients. The largest age group was [70-80), followed by [60-70), while relatively few encounters involved patients younger than 30 years. This indicates that the dataset is concentrated mainly among middle-aged and older adults.

4.4 Previous Inpatient Visits and Readmission

A particularly strong relationship was observed between the number of previous inpatient visits and 30-day readmission.

The readmission rate increased substantially as number_inpatient increased.

Previous inpatient visits 30-day readmission rate

0	8.4%
1	12.9%
2	17.4%
3	20.3%
4	23.6%
5	31.4%

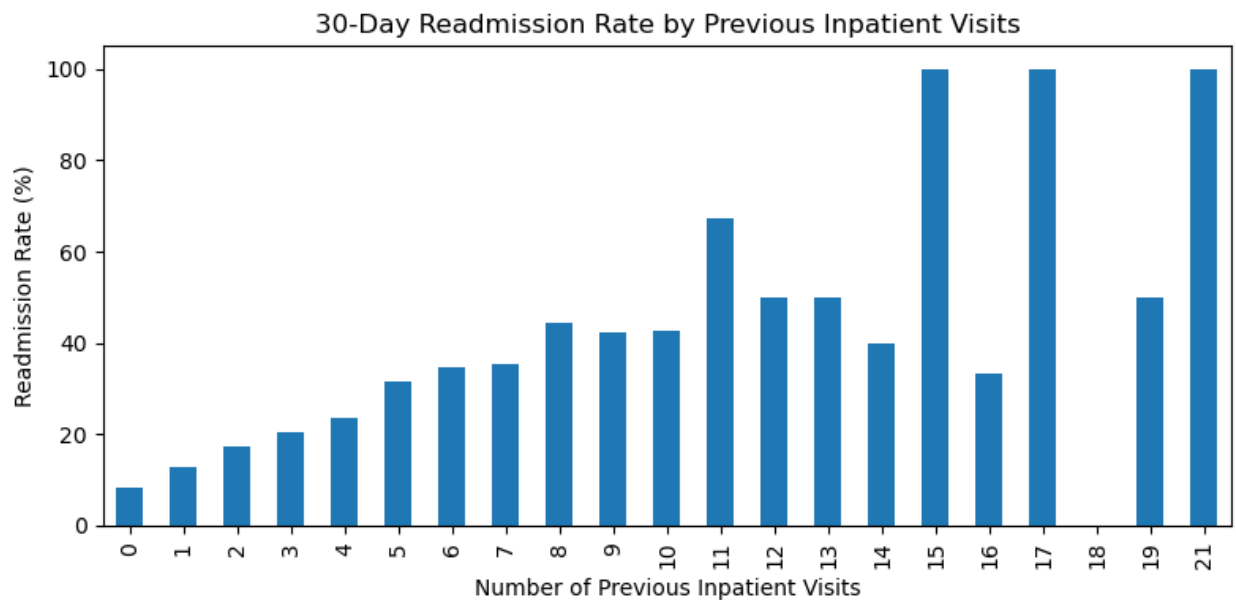


Figure 3

Figure 3 shows that the 30-day readmission rate generally increases as the number of previous inpatient visits increases. Patients with no previous inpatient visits had the lowest readmission rate, while patients with multiple previous inpatient visits showed substantially higher rates. Although some of the highest categories reached 100%, these were based on very small numbers of observations and should therefore be interpreted with caution.

4.5 Previous Emergency Visits and Readmission

A similar, although less regular, pattern was observed for previous emergency visits.

The readmission rate increased from approximately **10.5%** among patients with no previous emergency visits to approximately:

- 14.4% for one previous emergency visit;
- 18.3% for two visits;
- 20.3% for three visits; and
- 30.7% for four visits.

4.6 Length of Stay and Number of Medications

Longer hospital stays generally showed somewhat higher readmission rates, although the pattern was not perfectly monotonic.

The number of medications also showed some association with readmission. For exploratory purposes, medication counts were temporarily grouped into:

- 1–10
- 11–20
- 21–30
- 31–40
- 41+

The readmission rate increased from approximately **9.1%** for patients receiving 1–10 medications to around **12–13%** for several higher medication groups.

The grouped medication variable was used only for exploratory analysis; the original numerical `num_medications` variable was retained for modelling.

4.7 Diagnosis Group and Insulin Status

Readmission rates also differed across primary diagnosis groups.

Examples included:

- Diabetes — approximately 13.0%
- Injury — approximately 12.2%
- Circulatory — approximately 11.5%
- Respiratory — approximately 9.7%
- Musculoskeletal — approximately 9.5%

Insulin status also showed differences in observed readmission rates:

Insulin status 30-day readmission rate

No	10.0%
Steady	11.1%
Up	13.0%
Down	13.9%

4.8 Outliers and Unusual Patterns

Boxplots were used to investigate potential outliers in the numerical predictors.

Variables such as `num_lab_procedures`, `num_medications`, `number_outpatient`, `number_emergency`, and `number_inpatient` contained observations substantially above their typical ranges.

These observations were retained because they were considered clinically plausible and may represent high-utilisation or medically complex patients. Removing them solely on the basis of statistical outlier rules could remove information relevant to readmission risk.

A representative boxplot is presented in Figure 4, while the remaining boxplots are available in the Python notebook

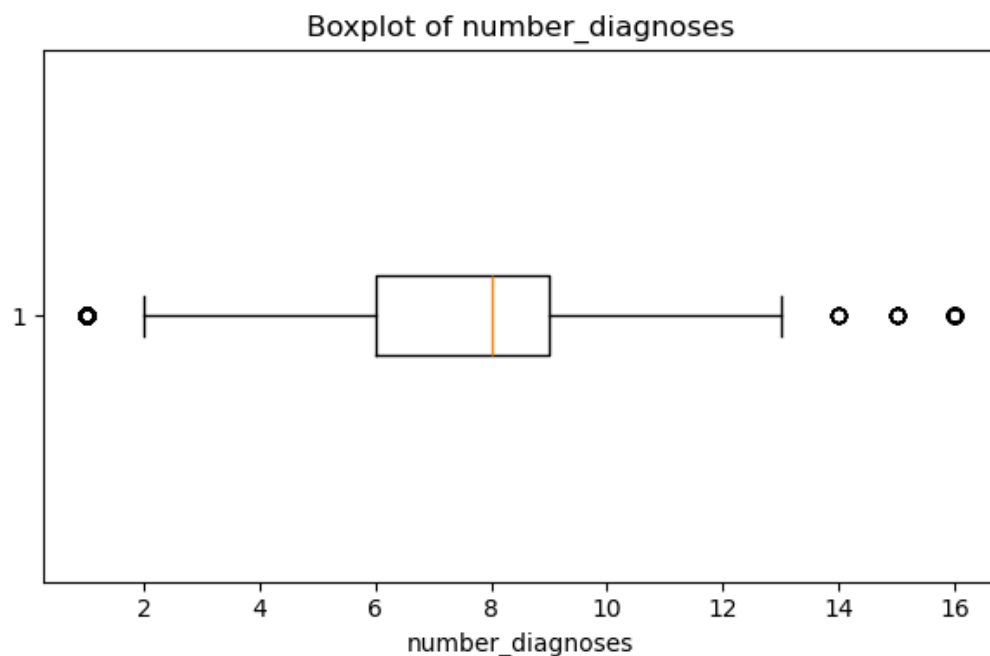


Figure 4

Figure 4 presents the boxplot for `number_diagnoses`. The distribution shows a median of about 8 diagnoses per encounter, with most observations lying between 6 and 9. A few outliers are visible at both the lower and upper ends, particularly for encounters with 14 to 16 diagnoses. These values were retained because they are likely to represent genuine complex patient cases rather than data errors, and such complexity may be relevant for predicting 30-day readmission.

4.9 Expected Important Predictors

Based on the exploratory analysis, two predictors were expected to be particularly important:

- `number_inpatient`
- `number_emergency`

`number_inpatient` showed the clearest and most consistent increase in readmission rate as previous inpatient utilisation increased.

`number_emergency` also showed a positive association with 30-day readmission, although the relationship was somewhat less consistent at higher values.

These expectations were later compared with the results from logistic regression and machine-learning models.

5. Logistic Regression Analysis

5.1 Full Binary Logistic Regression Model

A binary logistic regression model was fitted using the training data to predict `readmit_30`, where:

1 = Readmitted within 30 days

0 = Not readmitted within 30 days

Categorical predictors were converted into dummy variables using one-hot encoding, with one category omitted from each categorical variable as the reference category. After preprocessing and rare-category adjustments, the full model contained **96 encoded predictor terms**.

The logistic regression model converged successfully after the rare categorical levels were combined or removed. This was important because extremely sparse categories initially caused unstable coefficient estimates and convergence problems.

The overall likelihood-ratio test for the fitted model was statistically significant, indicating that the predictor set collectively provided information about 30-day readmission.

5.2 Most Statistically Significant Predictors

The three predictor terms with the smallest p-values in the full logistic regression model were:

Predictor term	Coefficient	p-value	Direction
number_inpatient	0.261	2.99×10^{-271}	Positive
discharge_group_Transfer/Facility	0.578	6.25×10^{-80}	Positive
diag_1_group_Respiratory	-0.255	1.66×10^{-10}	Negative

`number_inpatient` was the most statistically significant term in the model. Its positive coefficient indicates that encounters involving patients with more previous inpatient visits were associated with higher odds of 30-day readmission, after controlling for the other predictors.

`discharge_group_Transfer/Facility` also had a positive coefficient. The reference category for discharge group was **Home**, so this result indicates higher odds of 30-day readmission for encounters discharged to another facility compared with encounters discharged home.

diag_1_group_Respiratory had a negative coefficient. The reference category for the primary diagnosis group was **Circulatory**, so respiratory primary diagnoses were associated with lower odds of 30-day readmission relative to circulatory diagnoses, holding the other predictors constant.

5.3 Odds Ratio Interpretation

Because logistic regression coefficients are expressed on the log-odds scale, the coefficients were exponentiated to obtain odds ratios.

Predictor term	Odds ratio	Interpretation
number_inpatient	1.298	About 29.8% higher odds per additional previous inpatient visit
discharge_group_Transfer/Facility	1.783	About 78.3% higher odds than discharge Home
diag_1_group_Respiratory	0.775	About 22.5% lower odds than Circulatory diagnosis

5.4 Reduced Logistic Regression Model

A backward-elimination procedure was used to produce a more parsimonious logistic regression model.

The procedure began with the full model and sequentially removed the predictor term with the largest p-value greater than 0.05. The model was refitted after each removal until all retained terms met the selected significance criterion.

The number of predictor terms was reduced from:

Full model: 96

Reduced model: 45

The reduced model converged successfully.

This indicates that a substantial number of terms could be removed while retaining a statistically stable logistic regression model.

5.5 Full and Reduced Model Performance

The full and reduced models were evaluated on the untouched test data.

Metric	Full model	Reduced model
Accuracy	0.8866	0.8866
Precision	0.4921	0.4918
Recall	0.0137	0.0132
F1-score	0.0266	0.0257
ROC-AUC	0.6663	0.6654

The two models produced almost identical predictive performance. Therefore, the reduced model achieved similar predictive ability with substantially fewer predictor terms.

However, the very low recall at the default 0.50 threshold shows that both models identified very few actual 30-day readmissions.

5.6 Effect of the Classification Threshold

Because the outcome was highly imbalanced, the effect of changing the classification threshold was examined using the reduced logistic regression model.

Threshold	Accuracy	Precision	Recall	F1-score
0.10	0.5716	0.1645	0.6816	0.2651
0.15	0.7913	0.2206	0.3320	0.2651
0.20	0.8568	0.2746	0.1607	0.2028
0.30	0.8817	0.3621	0.0572	0.0989
0.50	0.8866	0.4918	0.0132	0.0257

Lowering the threshold substantially increased recall. At a threshold of 0.10, approximately **68.2%** of actual 30-day readmissions were identified, compared with only about **1.3%** at the default threshold of 0.50.

This improvement came at the cost of lower accuracy and precision because more non-readmitted encounters were classified as high risk.

5.7 Confusion Matrix Interpretation

At the default threshold of 0.50, the reduced model produced:

True negatives = 17,735

False positives = 31

False negatives = 2,241

True positives = 30

At the lower threshold of 0.10:

True negatives = 9,906

False positives = 7,860

False negatives = 723

True positives = 1,548

The lower threshold therefore identified many more true readmissions but also generated substantially more false-positive classifications.

5.8 Logistic Regression Assessment

Overall, the reduced logistic regression model provided nearly the same predictive performance as the full model while using fewer terms.

The ROC-AUC of approximately **0.665** indicates moderate discrimination, but classification performance was strongly dependent on the selected threshold.

For a readmission-screening application, a lower threshold may be useful when identifying high-risk encounters is more important than minimising false positives. The choice of threshold should therefore depend on the practical consequences of false-negative and false-positive classifications.

6. Machine-Learning Models

Two machine-learning classifiers were developed and evaluated: **Random Forest** and **Gradient Boosting**. These models were compared with logistic regression to determine whether nonlinear machine-learning approaches could improve prediction of 30-day readmission.

Because the response variable was imbalanced, model evaluation focused not only on accuracy but also on **precision, recall, F1-score, and ROC-AUC**.

6.1 Random Forest Classifier

A Random Forest classifier was first fitted using the encoded training data. Random Forest is an ensemble method that combines predictions from many decision trees.

The initial model used:

- 300 decision trees
- `class_weight="balanced"`
- `random_state=42`
- all available processor cores

Class weighting was used to give greater importance to the minority readmission class.

The initial Random Forest produced approximately:

Metric	Initial Random Forest
Accuracy	0.8868
Precision	0.5625
Recall	0.0040
F1-score	0.0079
ROC-AUC	0.6576

Although overall accuracy was high, recall was extremely low. This indicates that the initial Random Forest classified very few actual readmissions correctly.

6.2 Random Forest Hyperparameter Tuning

Randomized cross-validation was used to tune the Random Forest model.

The tuning process considered combinations of:

- number of trees;
- maximum tree depth;
- minimum samples required for a split;
- minimum samples per leaf; and
- number of predictors considered at each split.

ROC-AUC was used as the tuning criterion because the outcome was imbalanced.

The best parameter combination identified was:

`n_estimators = 500`

`max_depth = 30`

`min_samples_split = 2`

`min_samples_leaf = 5`

`max_features = "sqrt"`

The best cross-validation ROC-AUC was approximately **0.659**.

6.3 Tuned Random Forest Performance

The tuned Random Forest achieved the following test-set performance:

Metric	Tuned Random Forest
Accuracy	0.8649
Precision	0.2908
Recall	0.1334
F1-score	0.1829
ROC-AUC	0.6735

Tuning improved recall, F1-score, and ROC-AUC substantially compared with the initial Random Forest.

Recall increased from approximately **0.4% to 13.3%**, indicating better detection of actual readmissions.

However, the improvement in recall was accompanied by somewhat lower overall accuracy

6.4 Random Forest Overfitting Assessment

The tuned Random Forest achieved:

Training ROC-AUC = 0.990

Test ROC-AUC = 0.673

Difference = 0.316

The large difference between training and test ROC-AUC indicates substantial overfitting.

The model fitted the training data extremely well but did not generalise equally well to unseen observations.

6.5 Random Forest Variable Importance

Variable importance was examined using the tuned Random Forest model.

The most important predictor terms were:

Predictor	Importance
num_lab_procedures	0.0985
number_inpatient	0.0856
num_medications	0.0852
time_in_hospital	0.0620
number_diagnoses	0.0438

Other important variables included num_procedures, discharge_group_Transfer/Facility, gender_Male, number_outpatient, and number_emergency.

The importance of number_inpatient was consistent with both the exploratory analysis and logistic regression results.

Random Forest importance values indicate relative contribution to prediction and should not be interpreted as causal effects.

6.6 Gradient Boosting Classifier

A Histogram-based Gradient Boosting classifier was used as the second machine-learning model.

Unlike Random Forest, which constructs many trees independently, Gradient Boosting builds trees sequentially, with each new stage attempting to improve the errors made by previous stages.

The initial model used:

```
learning_rate = 0.10
```

```
max_iter = 200
```

```
max_leaf_nodes = 31
```

```
class_weight = "balanced"
```

The initial Gradient Boosting model achieved:

Metric	Initial Gradient Boosting
Accuracy	0.6519
Precision	0.1856
Recall	0.6116
F1-score	0.2848
ROC-AUC	0.6765

The most notable result was the substantially higher recall. The model correctly identified approximately **61.2% of actual 30-day readmissions**.

This came at the cost of lower accuracy and precision because more non-readmitted encounters were classified as being at risk.

6.7 Initial Gradient Boosting Overfitting Assessment

The initial model achieved:

Training ROC-AUC = 0.739

Test ROC-AUC = 0.677

Difference = 0.062

The relatively small gap indicates considerably less overfitting than the Random Forest model.

6.8 Gradient Boosting Hyperparameter Tuning

Randomized cross-validation was also used to tune the Gradient Boosting classifier.

The tuning process examined:

- learning rate;
- number of boosting iterations;
- maximum leaf nodes;
- minimum observations per leaf; and
- L2 regularisation.

The best parameter combination identified was:

min_samples_leaf = 50

max_leaf_nodes = 15

max_iter = 200

learning_rate = 0.05

l2_regularization = 0.1

The best cross-validation ROC-AUC was approximately **0.660**.

6.9 Tuned Gradient Boosting Performance

The tuned Gradient Boosting model achieved:

Metric	Tuned Gradient Boosting
Accuracy	0.6370
Precision	0.1806
Recall	0.6231
F1-score	0.2801
ROC-AUC	0.6772

Compared with the initial model, tuning produced a small improvement in recall and ROC-AUC while slightly reducing accuracy, precision, and F1-score.

Overall, the tuned and untuned Gradient Boosting models produced similar predictive performance

6.10 Tuned Gradient Boosting Overfitting Assessment

The tuned Gradient Boosting model achieved:

Training ROC-AUC = 0.700

Test ROC-AUC = 0.677

Difference = 0.023

The small difference suggests limited overfitting and substantially better generalisation than the tuned Random Forest.

6.11 Machine-Learning Model Summary

Overall, the tuned Random Forest achieved higher accuracy than Gradient Boosting but showed substantial overfitting and relatively low recall.

Gradient Boosting produced much higher recall and a slightly higher ROC-AUC while showing substantially less overfitting.

Therefore, among the two machine-learning classifiers, **Gradient Boosting provided the more balanced and generalisable performance for identifying 30-day readmissions.**

7. Model Comparison and Discussion

7.1 Comparison of Final Models

The final predictive performance of the **reduced logistic regression**, **tuned Random Forest**, and **tuned Gradient Boosting** models was compared using the same untouched test dataset.

Model	Accuracy	Precision	Recall	F1-score	ROC-AUC
Reduced Logistic Regression	0.8866	0.4918	0.0132	0.0257	0.6654
Tuned Random Forest	0.8649	0.2908	0.1334	0.1829	0.6735
Tuned Gradient Boosting	0.6370	0.1806	0.6231	0.2801	0.6772

The models showed substantially different trade-offs.

The reduced logistic regression model achieved the highest overall accuracy and precision at the default classification threshold of 0.50. However, its recall was extremely low, meaning that it failed to identify most actual 30-day readmissions.

The tuned Random Forest improved recall and F1-score compared with logistic regression, while maintaining relatively high accuracy. However, its recall remained low and the model showed substantial overfitting.

The tuned Gradient Boosting model achieved the highest recall, F1-score, and ROC-AUC. It therefore identified a much larger proportion of actual 30-day readmissions, although this was accompanied by lower accuracy and precision.

7.2 Recommended Model

For a readmission-risk screening application, the **tuned Gradient Boosting model** would be the preferred model among the three evaluated approaches.

The main reasons are:

- highest test ROC-AUC, approximately **0.677**;
- highest recall, approximately **62.3%**;
- highest F1-score, approximately **0.280**; and
- a relatively small training-test ROC-AUC gap, indicating limited overfitting.

In a healthcare screening context, failing to identify an encounter that is genuinely at high risk of readmission may be more important than incorrectly flagging some lower-risk encounters. Therefore, the higher recall of Gradient Boosting is practically useful.

However, its relatively low precision means that many encounters identified as high risk would not actually be readmitted within 30 days. The model should therefore be considered a **risk-screening or decision-support tool**, rather than a stand-alone clinical decision system.

7.3 Comparison with Exploratory Data Analysis

The exploratory analysis identified `number_inpatient` and `number_emergency` as two predictors expected to be important for 30-day readmission.

The modelling results strongly supported the importance of `number_inpatient`.

It was:

- associated with a clear increase in observed readmission rates during EDA;
- the most statistically significant term in the logistic regression model; and
- one of the most important predictors in the Random Forest model.

`number_emergency` also showed a positive association with readmission during EDA and was statistically significant in the logistic regression model. However, its relative importance in Random Forest was lower than several other predictors.

Therefore, the final models strongly supported the EDA expectation for `number_inpatient`, while the evidence for `number_emergency` was positive but less dominant.

7.4 Comparison of Important Predictors Across Models

The Random Forest and logistic regression models did not identify exactly the same variables as most important.

The main Random Forest predictor terms included:

- `num_lab_procedures`
- `number_inpatient`
- `num_medications`
- `time_in_hospital`
- `number_diagnoses`

The strongest logistic regression terms included:

- `number_inpatient`
- `discharge_group_Transfer/Facility`

- diag_1_group_Respiratory
- The strongest agreement between the two approaches was for number_inpatient.
- The differences are expected because the methods measure importance differently. Logistic regression evaluates the statistical association of individual predictor terms with the response while controlling for other variables. Random Forest importance reflects how much variables contribute to prediction across many decision trees.
- Therefore, different importance rankings do not necessarily indicate contradictory results.

7.5 Effect of Classification Threshold

The analysis also demonstrated that classification performance depends strongly on the selected decision threshold.

For the reduced logistic regression model, the default threshold of 0.50 produced:

- high overall accuracy;
- relatively high precision; but
- extremely low recall.

When the threshold was reduced to 0.10, recall increased from approximately **1.3% to 68.2%**.

However, the number of false-positive classifications also increased substantially.

This demonstrates that the threshold should not be chosen automatically. The appropriate value depends on the practical consequences of the two types of classification error:

- **False negative:** an encounter at genuine risk of readmission is not identified.
- **False positive:** an encounter is flagged as high risk but is not subsequently readmitted.

For a screening application, a lower threshold may be appropriate if the cost of missing a high-risk encounter is considered greater than the cost of additional follow-up for false-positive cases

7.6 Overall Discussion

The analysis shows that no single metric provides a complete assessment of model performance for this imbalanced classification problem.

The logistic regression model was highly interpretable and provided useful information about statistically significant predictors and odds ratios. However, at the default threshold it performed poorly in identifying actual readmissions.

Random Forest improved minority-class detection after tuning but showed substantial overfitting.

Gradient Boosting provided the strongest balance of discrimination, recall, F1-score, and generalisation among the models considered.

Overall, the results demonstrate the importance of considering **class imbalance, threshold selection, model interpretability, and generalisation** rather than relying on accuracy alone.

8. Limitations

This study has several limitations. The dataset is historical and contains hospital encounters from 1999 to 2008, so current clinical practice may differ. The same patient may appear in multiple encounters, meaning that observations may not be completely independent. The response variable was highly imbalanced, with only about 11% of encounters resulting in readmission within 30 days. Several variables also contained substantial missingness and required removal or grouping during preprocessing.

The predictive performance of the final models was moderate, and the Random Forest model showed substantial overfitting. In addition, classification performance was sensitive to the selected probability threshold. Finally, the observed relationships should be interpreted as associations rather than causal effects.

9. Conclusion

This study developed and compared statistical and machine-learning models for predicting **30-day hospital readmission** using the Diabetes 130-US Hospitals dataset.

The exploratory analysis showed that previous healthcare utilisation was strongly related to readmission risk. In particular, `number_inpatient` showed a clear increase in observed readmission rates and later emerged as one of the strongest predictors across the modelling approaches.

The logistic regression analysis provided useful interpretability through coefficients, p-values, and odds ratios. `number_inpatient` was the most statistically significant predictor term, and the reduced logistic model achieved almost the same predictive performance as the full model while using substantially fewer predictor terms. However, at the default 0.50 threshold, recall was extremely low because of the strong class imbalance.

The tuned Random Forest improved recall and ROC-AUC compared with the initial Random Forest model, but it showed substantial overfitting, with a large gap between training and test ROC-AUC.

The tuned Gradient Boosting model produced the strongest overall predictive balance. It achieved the highest recall, F1-score, and ROC-AUC among the final models while showing only a small difference between training and test ROC-AUC. This indicated better generalisation to unseen data.

Therefore, the tuned Gradient Boosting model was considered the most suitable of the evaluated models for a **readmission-risk screening application**. However, its relatively low precision means that it would generate a substantial number of false-positive classifications. For this reason, the model should be used as a decision-support tool rather than as a stand-alone clinical decision system.

The analysis also demonstrated that classification threshold selection is important in imbalanced healthcare prediction problems. Lower thresholds substantially improved recall but increased false positives. Therefore, the final operating threshold should depend on the practical cost of missing a genuinely high-risk encounter compared with the cost of additional follow-up for a false-positive case.

Overall, the study showed that combining exploratory analysis, interpretable statistical modelling, and machine-learning approaches provides a useful framework for understanding and predicting 30-day hospital readmission.

10. References and Data Source

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Data Source Note

The main analytical dataset was obtained from the UCI Machine Learning Repository. The accompanying IDS_mapping.csv file supplied with the dataset was used to interpret and group coded variables including admission type, discharge disposition, and admission source.

Use of Generative AI

ChatGPT by OpenAI was used as a supporting tool for code explanation, interpretation of modelling outputs, report structuring, and drafting assistance. All Python analyses were executed using the submitted notebook, and the analytical results and final interpretations were reviewed against the generated outputs.

11. Code Availability

The complete Python analysis is provided in the submitted Jupyter Notebook:

24ada006_Code.ipynb

The project repository, including the analysis workflow and supporting files, is also available on GitHub as a supplementary reference:.

https://github.com/ashokaw/DA3014_Assignment3